

CSA 0378-DATA STRUCTURES

10. STACK USING ARRAY

```
main.c
1 #include <stdio.h>
2
3 #define MAX 5
4
5 int stack[MAX];
6 int top = -1;
7
8 void push(int value)
9 {
10     if(top == MAX - 1)
11     {
12         printf("Stack Overflow\n");
13         return;
14     }
15     stack[++top] = value;
16     printf("%d pushed into stack\n", value);
17 }
18
19 void pop()
20 {
21     if(top == -1)
22     {
23         printf("Stack Underflow\n");
24         return;
25     }
26     printf("Popped Element: %d\n", stack[top--]);
27 }
28
29 void display()
30 {
31     int i;
32     if(top == -1)
33     {
34         printf("Stack is Empty\n");
35         return;
36     }
```

```
Output
1. Push
2. Pop
3. Display
4. Exit
Enter Choice: 1
Enter Value: 2
2 pushed into stack

1. Push
2. Pop
3. Display
4. Exit
Enter Choice: 1
Enter Value: 3
3 pushed into stack

1. Push
2. Pop
3. Display
4. Exit
Enter Choice: 2
Popped Element: 3

1. Push
2. Pop
3. Display
4. Exit
Enter Choice: 2
Popped Element: 2

1. Push
2. Pop
3. Display
4. Exit
Enter Choice:
```

```
main.c
31 void display()
32 {
33     int i;
34     if(top == -1)
35     {
36         printf("Stack is Empty\n");
37         return;
38     }
39     printf("Stack Elements:\n");
40     for(i = top; i >= 0; i--)
41         printf("%d ", stack[i]);
42     printf("\n");
43 }
44
45 int main()
46 {
47     int choice, value;
48     while(1)
49     {
50         printf("\n1. Push\n");
51         printf("2. Pop\n");
52         printf("3. Display\n");
53         printf("4. Exit\n");
54         printf("Enter Choice: ");
55         scanf("%d", &choice);
56
57         if(choice == 1)
58         {
59             printf("Enter Value: ");
60             scanf("%d", &value);
61             push(value);
62         }
```

```
Output
1. Push
2. Pop
3. Display
4. Exit
Enter Choice: 1
Enter Value: 2
2 pushed into stack

1. Push
2. Pop
3. Display
4. Exit
Enter Choice: 1
Enter Value: 3
3 pushed into stack

1. Push
2. Pop
3. Display
4. Exit
Enter Choice: 2
Popped Element: 3

1. Push
2. Pop
3. Display
4. Exit
Enter Choice: 2
Popped Element: 2

1. Push
2. Pop
3. Display
4. Exit
Enter Choice:
```

```
main.c
52 while(1)
53 {
54     printf("\n1. Push\n");
55     printf("2. Pop\n");
56     printf("3. Display\n");
57     printf("4. Exit\n");
58     printf("Enter Choice: ");
59     scanf("%d", &choice);
60
61     if(choice == 1)
62     {
63         printf("Enter Value: ");
64         scanf("%d", &value);
65         push(value);
66     }
67     else if(choice == 2)
68     {
69         pop();
70     }
71     else if(choice == 3)
72     {
73         display();
74     }
75     else if(choice == 4)
76     {
77         break;
78     }
79     else
80     {
81         printf("Invalid Choice\n");
82     }
83 }
84
85 return 0;
86 }
```

```
Output
1. Push
2. Pop
3. Display
4. Exit
Enter Choice: 1
Enter Value: 2
2 pushed into stack

1. Push
2. Pop
3. Display
4. Exit
Enter Choice: 1
Enter Value: 3
3 pushed into stack

1. Push
2. Pop
3. Display
4. Exit
Enter Choice: 2
Popped Element: 3

1. Push
2. Pop
3. Display
4. Exit
Enter Choice: 2
Popped Element: 2

1. Push
2. Pop
3. Display
4. Exit
Enter Choice:
```